according to the transport ray containing the point x. More precisely, we define a map $R:\Omega\times\Omega\to\mathscr{R}$, valued in the set $\mathscr{R}$ of all transport rays, sending each pair (x,y) into the ray containing x. This is well-defined γ-a.e. and we can write $\gamma=\gamma^r\otimes\lambda$, where $\lambda=R_\#\gamma$ and we denote by r the variable related to transport rays. Note that, for a.e. $r\in\mathscr{R}$, the plan γ^r is optimal between its own marginals (otherwise we could replace it with an optimal plan, do it in a measurable way, and improve the cost of γ).

The measure $\mathbf{w}_{[\gamma]}$ may also be obtained through this disintegration, and we have $\mathbf{w}_{[\gamma]}=\mathbf{w}_{\gamma^r}\otimes\lambda$. This means that, in order to prove that $\mathbf{w}_{[\gamma]}$ does not depend on γ, we just need to prove that each $\mathbf{w}_{\gamma^r}$ and the measure λ do not depend on it. For the measure λ this is easy: it has been obtained as an image measure through a map only depending on x, and hence only depends on μ. Concerning $\mathbf{w}_{\gamma^r}$, note that it is obtained in the standard Beckmann way from an optimal plan, γ^r. Hence, thanks to the considerations in Section 4.2.4 about the 1D case, it uniquely depends on the marginal measures of this plan.

This means that we only need to prove that $(\pi_x)_\#\gamma^r$ and $(\pi_y)_\#\gamma^r$ do not depend on γ. Again, this is easy for $(\pi_x)_\#\gamma^r$, since it must coincide with the disintegration of μ according to the map R (by uniqueness of the disintegration). It is more delicate for the second marginal.

The second marginal $\nu^r:=(\pi_y)_\#\gamma^r$ will be decomposed into two parts:

$$(\pi_y)_\#(\gamma^r_{|\Omega\times S})\;+\;(\pi_y)_\#(\gamma^r_{|\Omega\times S^c}).$$

This second part coincides with the disintegration of $\nu_{|S^c}$, which obviously does not depend on γ (since it only depends on the set S, which is built upon u).

We need now to prove that $\nu^r_{|S}=(\pi_y)_\#(\gamma^r_{|\Omega\times S})$ does not depend on γ. Yet, this measure can only be concentrated on the two endpoints of the transport ray r, since these are the only points where different transport rays can meet. This means that this measure is purely atomic and composed by at most two Dirac masses. Not only, the endpoint where u is maximal (i.e. the first one for the order on the ray, see Section 3.1) cannot contain some mass of ν: indeed the transport must follow a precise direction on each transport ray (as a consequence of $u(x)-u(y)=|x-y|$ on $\operatorname{spt}(\gamma)$), and the only way to have some mass of the target measure at the "beginning" of the transport ray would be to have an atom for the source measure as well. Yet, μ is absolutely continuous and Property N holds (see Section 3.1.4 and Theorem 3.12, which means that the set of rays r where μ^r has an atom is negligible). Hence $\nu^r_{|S}$ is a single Dirac mass. The mass equilibrium condition between μ^r and ν^r implies that the value of this mass must be equal to the difference $1-\nu^r_{|S^c}(r)$, and this last quantity does not depend on γ but only on μ and ν.

Finally, this proves that each $\mathbf{w}_{\gamma^r}$ does not depend on the choice of γ. □ □

Corollary 4.15. *If $\mu\ll\mathscr{L}^d$, then the solution of* (BP) *is unique.*

Proof. We have seen in Theorem 4.13 that any optimal $\mathbf{w}$ is of the form $\mathbf{w}_{[\gamma]}$ and in Theorem 4.14 that all the fields $\mathbf{w}_{[\gamma]}$ coincide. □ □

4.3 Summability of the transport density

The analysis of the Beckmann Problem performed in the previous sections was mainly made in a measure setting, and the optimal $\mathbf{w}$, as well as the transport density σ, were just measures on Ω. We investigate here the question whether they have extra regularity properties supposing extra assumptions on μ and/or ν.

We will give summability results, proving that σ is in some cases absolutely continuous and providing L^p estimates. The proofs are essentially taken from [273]: previous results, through very different techniques, were first presented in [144, 146, 145]. In these papers, different estimates on the "dimension" of σ were also presented, thus giving interesting information should σ fail to be absolutely continuous. Let us also observe that [209], while applying these tools to image processing problems, contains a new[4] proof of absolute continuity.

Note that higher order questions (such as whether σ is continuous or Lipschitz or more regular provided μ and ν have smooth densities) are completely open up to now. The only exception is a partial result in dimension 2, see [172], where a continuity result is given if μ and ν have Lipschitz densities on disjoint convex domains.

Open Problem (continuity of the transport density): is it true that, supposing that μ,ν have continuous densities, the transport density σ is also continuous? is it true that it is Lipschitz if they are Lipschitz, and/or Hölder if they are Hölder?

In all that follows Ω is a compact and convex domain in $\mathbb{R}^d$, and two probability measures μ,ν are given on it. At least one of them will be absolutely continuous, which implies uniqueness for σ (see Theorem 4.14).

Theorem 4.16. *Suppose $\Omega \subset \mathbb{R}^d$ is a convex domain, and $\mu \ll \mathscr{L}^d$. Let σ be the transport density associated to the transport of μ onto ν. Then $\sigma \ll \mathscr{L}^d$.*

Proof. Let γ be an optimal transport plan from μ to ν and take $\sigma = \sigma_\gamma$; call μ_t the standard interpolation between the two measures: $\mu_t = (\pi_t)_\#\gamma$ where $\pi_t(x,y) = (1-t)x+ty$: we have $\mu_0 = \mu$ and $\mu_1 = \nu$.

We have already seen (go back to (4.6)) that the transport density σ may be written as
$$\sigma = \int_0^1 (\pi_t)_\#(c\cdot\gamma)\mathrm{d}t,$$
where $c : \Omega\times\Omega \to \mathbb{R}$ is the cost function $c(x,y) = |x-y|$ (hence $c\cdot\gamma$ is a positive measure on $\Omega\times\Omega$).

Since Ω is bounded it is evident that we have
$$\sigma \le C\int_0^1 \mu_t\,\mathrm{d}t. \tag{4.11}$$

To prove that σ is absolutely continuous, it is sufficient to prove that almost every measure μ_t is absolutely continuous, so that, whenever $|A| = 0$, we have $\sigma(A) \le C\int_0^1 \mu_t(A)\mathrm{d}t = 0$.

[4] Their proof is somehow intermediate between that of [144] and the one we present here: indeed, approximation by atomic measures is also performed in [209], as here, but on both the source and the target measure, which requires a geometric analysis of the transport rays as in [144].

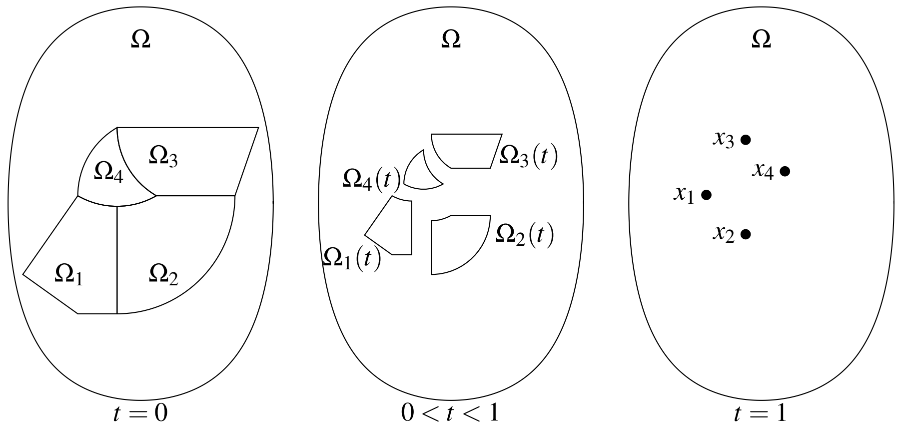

Figure 4.2: Disjointness of the sets $\Omega_i(t)$

We will prove $\mu_t \ll \mathscr{L}^d$ for $t < 1$. First, we will suppose that ν is finitely atomic (the point $(x_i)_{i=1,\dots,N}$ being its atoms). In this case we will choose γ to be any optimal transport plan induced by a transport map T (which exists, since $\mu \ll \mathscr{L}^d$). Note that the absolute continuity of σ is an easy consequence of the behavior of the optimal transport from μ to ν (which is composed by N homotheties), but we also want to quantify this absolute continuity, in order to go on with an approximation procedure.

Remember that μ is absolutely continuous and hence there exists a correspondence $\varepsilon \mapsto \delta = \delta(\varepsilon)$ such that

$$|A| < \delta(\varepsilon) \Rightarrow \mu(A) < \varepsilon. \tag{4.12}$$

Take now a Borel set A and look at $\mu_t(A)$. The domain Ω is the disjoint union of a finite number of sets $\Omega_i = \mathrm{T}^{-1}(\{x_i\})$. We call $\Omega_i(t)$ the images of Ω_i through the map $x \mapsto (1-t)x + t\mathrm{T}(x)$. These sets are essentially disjoint. Why? Because if a point z belongs to $\Omega_i(t)$ and $\Omega_j(t)$, then two transport rays cross at z, the one going from $x_i' \in \Omega_i$ to x_i and the one from $x_j' \in \Omega_j$ to x_j. The only possibility is that these two rays are actually the same, i.e. that the five points x_i', x_j', z, x_i, x_j are aligned. But this implies that z belongs to one of the lines connecting two atoms x_i and x_j. Since we have finitely many of these lines this set is negligible. Note that this argument only works for $d > 1$ (we will not waste time on the case $d = 1$, since the transport density is always a BV, and hence bounded, function). Moreover, if we sticked to the ray-monotone optimal transport map, we could have actually proved that these sets are truly disjoint, with no negligible intersection.

Now we have

$$\mu_t(A) = \sum_i \mu_t(A \cap \Omega_i(t)) = \sum_i \mu_0\left(\frac{A \cap \Omega_i(t) - tx_i}{1-t}\right) = \mu_0\left(\bigcup_i \frac{A \cap \Omega_i(t) - tx_i}{1-t}\right).$$

Since for every i we have

$$\left|\frac{A\cap\Omega_i(t)-tx_i}{1-t}\right| = \frac{1}{(1-t)^d}\,|A\cap\Omega_i(t)|$$

we obtain

$$\left|\bigcup_i \frac{A\cap\Omega_i(t)-tx_i}{1-t}\right| \leq \frac{1}{(1-t)^d}|A|.$$

Hence it is sufficient to suppose $|A| < (1-t)^d\delta(\varepsilon)$ to get $\mu_t(A) < \varepsilon$. This confirms $\mu_t \ll \mathscr{L}^d$ and gives an estimate that may pass to the limit.

Take a sequence ν^n of atomic measures converging to ν. The corresponding optimal transport plans γ^n converge to an optimal transport plan γ and μ_t^n converge to the corresponding μ_t (see Theorem 1.50 in Chapter 1). Hence, to prove absolute continuity for the transport density σ associated to such a γ it is sufficient to prove that these μ_t are absolutely continuous.

Take a set A such that $|A| < (1-t)^d\delta(\varepsilon)$. Since the Lebesgue measure is regular (see the definition in the memo Box 1.6), A is included in an open set B such that $|B| < (1-t)^d\delta(\varepsilon)$. Hence $\mu_t^n(B) < \varepsilon$. Passing to the limit, thanks to weak convergence and semi-continuity on open sets, we have

$$\mu_t(A) \leq \mu_t(B) \leq \liminf_n \mu_t^n(B) \leq \varepsilon.$$

This proves $\mu_t \ll \mathscr{L}^d$ and hence $\sigma \ll \mathscr{L}^d$. □ □

Remark 4.17. Where did we use the optimality of γ? We did it when we said that the $\Omega_i(t)$ are disjoint. For a discrete measure ν, it is always true that the measures μ_t corresponding to any transport plan γ are absolutely continuous for $t < 1$, but their absolute continuity may degenerate at the limit if we allow the sets $\Omega_i(t)$ to superpose (since in this case densities sum up and the estimates may depend on the number of atoms).

Remark 4.18. Note that we strongly used the equivalence between the two different definitions of absolute continuity, i.e. the $\varepsilon \leftrightarrow \delta$ correspondence on the one hand and the condition on negligible sets on the other. Indeed, to prove that the condition $\mu_t \ll \mathscr{L}^d$ passes to the limit we need the first one, while to deduce $\sigma \ll \mathscr{L}^d$ we need the second one.

Remark 4.19. As a byproduct of our proof we can see that any optimal transport plan from μ to ν which is approximable through optimal transport plans from μ to atomic measures must be such that all the interpolating measures μ_t (for *every* $t \in]0,1[$) are absolutely continuous. This property is not satisfied by any optimal transport plan, since for instance the plan γ which sends $\mu = \mathscr{L}^2_{|[-2,-1]\times[0,1]}$ onto $\nu = \mathscr{L}^2_{|[1,2]\times[0,1]}$ moving (x,y) to $(-x,y)$ is optimal but is such that $\mu_{1/2} = \mathscr{H}^1_{|\{0\}\times[0,1]}$. Hence, this plan cannot be approximated by optimal plans sending μ onto atomic measures. On the other hand, we proved in Lemma 3.20 that the monotone optimal transport can indeed be approximated in a similar way.

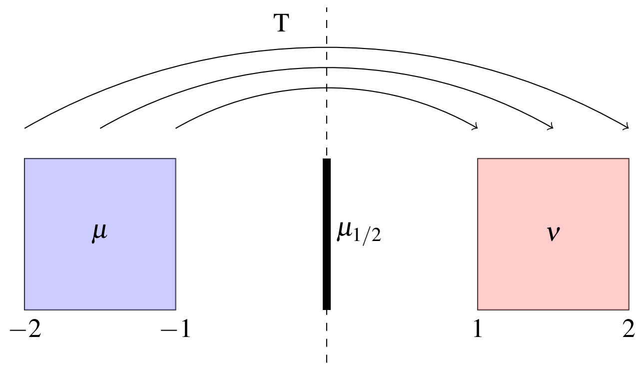

Figure 4.3: An example of optimal transport plan that cannot be approximated with optimal plans with atomic second marginal.

From now on we will often confuse absolutely continuous measures with their densities and write $||\mu||_{L^p}$ for $||f||_{L^p(\Omega)}$ when $\mu = f\cdot\mathscr{L}^d$.

Theorem 4.20. *Suppose $\mu = f\cdot\mathscr{L}^d$, with $f\in L^p(\Omega)$, where Ω is compact and convex. Then, if $p<d' := d/(d-1)$, the unique transport density σ associated to the transport of μ onto ν belongs to $L^p(\Omega)$ as well, and if $p\geq d'$ it belongs to any space $L^q(\Omega)$ for $q<d'$.*

Proof. Start from the case $p<d'$: following the same strategy (and the same notations) as before, it is sufficient to prove that each measure μ_t (for $t\in[0,1[$) is in L^p and to estimate their L^p norm. Then we will use

$$||\sigma||_{L^p}\leq C\int_0^1 ||\mu_t||_{L^p}\,\mathrm{d}t,$$

(which is a consequence of (4.11) and of Minkowski inequality), the conditions on p being chosen exactly so that this integral converges.

Consider first the discrete case: we know that μ_t is absolutely continuous and that its density coincides on each set $\Omega_i(t)$ with the density of an homothetic image of μ on Ω_i, the homothetic ratio being $(1-t)$. Hence, if f_t is the density of μ_t, we have

$$\begin{aligned}\int_\Omega f_t(x)^p\,\mathrm{d}x &= \sum_i\int_{\Omega_i(t)} f_t(x)^p\,\mathrm{d}x = \sum_i\int_{\Omega_i}\left(\frac{f(x)}{(1-t)^d}\right)^p(1-t)^d\,\mathrm{d}x\\ &= (1-t)^{d(1-p)}\sum_i\int_{\Omega_i} f(x)^p\,\mathrm{d}x = (1-t)^{d(1-p)}\int_\Omega f(x)^p\,\mathrm{d}x.\end{aligned}$$

We get $||\mu_t||_{L^p} = (1-t)^{-d/p'}||\mu||_{L^p}$, where $p'=p/(p-1)$.

This inequality, which is true in the discrete case, stays true at the limit as well. If ν is not atomic, approximate it through a sequence μ_1^n and take optimal plans γ^n and interpolating measures μ_t^n. Up to subsequences we have $\gamma^n\rightharpoonup\gamma$ (for an optimal transport plan γ) and $\mu_t^n \rightharpoonup \mu_t$ (for the corresponding interpolation); by semi-continuity we have

$$||\mu_t||_{L^p} \leq \liminf_n ||\mu_t^n||_{L^p} \leq (1-t)^{-d/p'}||\mu_0||_{L^p}$$

and we deduce

$$||\sigma||_{L^p} \leq C\int_0^1 ||\mu_t||_{L^p}\mathrm{d}t \leq C||\mu_0||_{L^p}\int_0^1 (1-t)^{-d/p'}\mathrm{d}t.$$

The last integral is finite whenever $p' > d$, i.e. $p < d' = d/(d-1)$.

The second part of the statement (the case $p \geq d'$) is straightforward once one considers that any density in L^p also belongs to any L^q space for $q < p$. □ □

Example 4.21. We can see an example where, even if $\mu \in L^\infty$, the singularity of ν prevents σ from being better than $L^{d'}$. Consider $\mu = f\cdot\mathscr{L}^d$ with $f = \mathbb{1}_A$ and $A = B(0,R)\setminus B(0,R/2)$ (with R chosen so that $\int f = 1$), and take $\nu = \delta_0$. Note that $\frac{R}{2}\gamma \leq c\cdot\gamma \leq R\gamma$, which implies that the summability of $\sigma = \int_0^1(\pi_t)_\#(c\cdot\gamma)\mathrm{d}t$, is the same as that of $\int_0^1(\pi_t)_\#(\gamma)\mathrm{d}t = \int_0^1\mu_t\,\mathrm{d}t$. We have $\mu_t = f_t\cdot\mathscr{L}^d$, with $f_t = (1-t)^{-d}\mathbb{1}_{(1-t)A}$, hence

$$\int_0^1 f_t(x)\,\mathrm{d}t = \int_{\frac{1}{R}|x|}^{\frac{2}{R}|x|}\frac{1}{s^d}\mathrm{d}s = c|x|^{1-d} \quad \text{for } |x| \leq \frac{R}{2}$$

(where we used the change of variable $s = 1-t$). This function belongs to L^p in a neighborhood of 0 in dimension d only if we have $(1-d)p + (d-1) > -1$, i.e. if $(d-1)(p-1) < 1$. This is exactly $p < d'$.

We saw in the previous theorems that the measures μ_t inherit some regularity (absolute continuity or L^p summability) from μ exactly as it happens for homotheties of ratio $1-t$. This regularity degenerates as $t \to 1$, but we saw two cases where this degeneracy produced no problem: for proving absolute continuity, where the separate absolute continuous behavior of almost all the μ_t was sufficient, and for L^p estimates, provided the degeneracy stays integrable.

It is natural to try to exploit another strategy: suppose both μ and ν share some regularity assumption (e.g., they belong to L^p). Then we can give estimate on μ_t for $t \leq 1/2$ starting from μ and for $t \geq 1/2$ starting from ν. In this way we have no degeneracy!

This strategy works quite well, but it has an extra difficulty: in our previous estimates we didn't know a priori that μ_t shared the same behavior of piecewise homotheties of μ, we got it as a limit from discrete approximations. And, when we pass to the limit, we do not know which optimal transport γ will be selected as a limit of the optimal plans γ^n. This was not important in the previous section, since any optimal γ induces the same transport density σ. But here we would like to glue together estimates on μ_t for $t \leq 1/2$ which have been obtained by approximating μ_1, and estimates on μ_t for $t \geq 1/2$ which come from the approximation of μ_0. Should the two approximations converge to two different transport plans, we could not put together the two estimates and deduce anything on σ.

Hence, the main technical issue that we need to consider is proving that one particular optimal transport plan, i.e. the ray-monotone one, can be approximated in both directions. Lemma 3.20 exactly does the job (and, indeed, it was proven in [273] exactly for this purpose). Yet, the transport plans γ_ε we build in the approximation are not optimal for the cost $|x-y|$ but for some costs $|x-y|+\varepsilon|x-y|^2$. We need to do this in order to force the selected limit optimal transport to be the monotone one (through a secondary variational problem, say). Anyway, this will not be an issue: these approximating optimal transport plans will share the same geometric properties that will imply disjointness for the sets $\Omega_i(t)$. In particular, we can prove the following estimate.

Lemma 4.22. *Let γ be an optimal transport plan between μ and an atomic measure ν for a transport cost $c(x,y)=h(y-x)$ where $h:\mathbb{R}^d\to\mathbb{R}$ is a strictly convex function. Set as usual $\mu_t=(\pi_t)_\#\gamma$. Then we have $||\mu_t||_{L^p}\leq(1-t)^{-d/p'}||\mu||_{L^p}$.*

Proof. The result is exactly the same as in Theorem 4.20, where the key tool is the fact that μ_t coincides on every set $\Omega_i(t)$ with a homothety of μ_0. The only fact that must be checked again is the disjointness of the sets $\Omega_i(t)$.

To do so, take a point $x\in\Omega_i(t)\cap\Omega_j(t)$. Hence there exist x_i,x_j belonging to Ω_i and Ω_j, respectively, so that $x=(1-t)x_i+ty_i=(1-t)x_j+ty_j$, being $y_i=\mathrm{T}(x_i)$ and $y_j=\mathrm{T}(x_j)$ two atoms of ν. But this would mean that $\mathrm{T}_t:=(1-t)\mathrm{id}+t\mathrm{T}$ is not injective, which is a contradiction to the following Lemma 4.23. Hence the sets $\Omega_i(t)$ are disjoint and this implies the bound on μ_t. □ □

Lemma 4.23. *Let γ be an optimal transport plan between μ and ν for a transport cost $c(x,y)=h(y-x)$ where $h:\mathbb{R}^d\to\mathbb{R}$ is a strictly convex function, and suppose that it is induced by a transport map* T. *Choose a representative of* T *such that $(x,\mathrm{T}(x))\in\mathrm{spt}(\gamma)$ for all x. Then the map $x\mapsto(1-t)x+t\mathrm{T}(x)$ is injective for $t\in]0,1[$.*

Proof. Suppose that there exist $x\neq x'$ such that

$$(1-t)x+t\mathrm{T}(x)=(1-t)x'+t\mathrm{T}(x')=x_0.$$

Set $a=\mathrm{T}(x)-x$ and $b=\mathrm{T}(x')-x'$. This also means $x=x_0-ta$ and $x'=x_0-tb$. In particular, $x\neq x'$ implies $a\neq b$.

The c-cyclical monotonicity of the support of the optimal γ implies

$$h(a)+h(b)\leq h(\mathrm{T}(x')-x)+h(\mathrm{T}(x)-x')=h(tb+(1-t)a)+h(ta+(1-t)b).$$

Yet, $a\neq b$, and strict convexity imply

$$h(tb+(1-t)a)+h(ta+(1-t)b)<th(b)+(1-t)h(a)+th(a)+(1-t)h(b)=h(a)+h(b),$$

which is a contradiction. □ □

Theorem 4.24. *Suppose that μ and ν are probability measures on Ω, both belonging to $L^p(\Omega)$, and σ the unique transport density associated to the transport of μ onto ν. Then σ belongs to $L^p(\Omega)$ as well.*

Proof. Let us consider the optimal transport plan $\overline{\gamma}$ from μ to ν defined by (3.4). We know that this transport plan may be approximated by plans γ^ε which are optimal for the cost $|x-y|+\varepsilon|x-y|^2$ from μ to some discrete atomic measures ν_ε. The corresponding interpolation measures μ_t^ε satisfy the L^p estimate from Lemma 4.22 and, at the limit, we have

$$||\mu_t||_{L^p} \leq \liminf_{\varepsilon\to 0} ||\mu_t^\varepsilon||_{L^p} \leq (1-t)^{-d/p'}||\mu||_{L^p}.$$

The same estimate may be performed from the other direction, since the very same transport plan $\overline{\gamma}$ may be approximated by optimal plans for the cost $|x-y|+\varepsilon|x-y|^2$ from atomic measures to ν. Putting together the two estimates we have

$$||\mu_t||_{L^p} \leq \min\left\{(1-t)^{-d/p'}||\mu||_{L^p}, t^{-d/p'}||\nu||_{L^p}\right\} \leq 2^{d/p'} \max\left\{||\mu||_{L^p}, ||\nu||_{L^p}\right\}.$$

Integrating these L^p norms we get the bound on $||\sigma||_{L^p}$. □ □

Remark 4.25. The same result could have been obtained in a strongly different way, thanks to the displacement convexity of the functional $\mu \mapsto ||\mu||_{L^p}^p$. This functional is actually convex along geodesics in the space $W_q(\Omega)$ for $q>1$ (see Proposition 7.29 where, unfortunately, the notation for p and q is reversed). Then we pass to the limit as $q\to 1$: this gives the result on the interpolating measures corresponding to the optimal plan which is obtained as a limit of the K_q-optimal plans. This plan is, by the way, $\overline{\gamma}$ again. And the integral estimate comes straightforward. Yet, this requires more sophisticated tools than what developed so far.

Example 4.26. We can provide examples (in dimension higher than one) where the summability of σ is no more than that of μ and ν. Take for instance $\mu = f\cdot\mathscr{L}^d$ and $\nu = g\cdot\mathscr{L}^d$ with $f(x_1,x_2,\dots,x_d) = f_0(x_2,\dots x_d)\mathbb{1}_{[0,1]}(x_1)$ and $g(x_1,x_2,\dots,x_d) = f_0(x_2,\dots x_d)\mathbb{1}_{[4,5]}(x_1)$. It is not difficult to check that the transport rays in this example go in the direction of the first coordinate vector, and that the optimal σ is absolutely continuous with density given by $\sigma(x) = f_0(x_2,\dots x_d)s(x_1)$, where $s:\mathbb{R}\to\mathbb{R}_+$ is characterized by $s' = \mathbb{1}_{[0,1]} - \mathbb{1}_{[4,5]}$, $s(0)=s(5)=0$ (which comes from the fact that s would be the transport density for the one-dimensional transport between $\mathbb{1}_{[0,1]}$ and $\mathbb{1}_{[4,5]}$). Hence, from $s=1$ on $[1,4]$, one sees that the summability of σ is the same of that of f_0, which is also the same of those of f and g.

4.4 Discussion

4.4.1 Congested transport

As we saw in Section 4.2, Beckmann's problem can admit an easy variant if we prescribe a positive function $k:\Omega\to\mathbb{R}_+$, where $k(x)$ stands for the local cost at x per unit length of a path passing through x. This models the possibility that the metric is non-homogeneous, due to geographical obstacles given a priori. Yet, it happens in many situations, in particular in urban traffic (as many readers probably note every day) that this metric k is indeed non-homogeneous, but is not given a priori: it depends on the traffic itself. In Beckmann's language, we must look for a vector field $\mathbf{w}$ optimizing a transport cost depending on $\mathbf{w}$ itself!